Game Design Document (GDD)

The LAB first game is an endless runner game that belongs to Skill-based game genre. It is a simple and cartoons-style 2D game. Figure 1 shows the game to be created.



Figure 1: LAB 2D game.

Game Overview

The player controls a small bird that moves forward automatically. Tapping the space bar in the keyboard makes it flap upward, and when he does not tap, it falls. The main job is to guide it through gaps between endless pipes. Each pair of pipes the player pass earns one point, and touching a pipe or the ground ends the game immediately.

Game in Depth

In this game, we have characters, user interface components, scenes, and sound effects, as follows.

- **Scenes**: the game consists of 2 scenes: Main and Welcome.
    - **Welcome Scene**: It is a welcoming screen when the player runs the game. It has the name of the game and two buttons. The first button is “Play” to begin the game by moving to the Main scene. The second button is “Exit” to terminate the game and close the window. Figure 2 shows the Welcome Scene.



Figure 2: The Welcome Scene.

- **Main Scene**: It is the main scene to play the game with all characters and game logic, as shown in Figure 3.

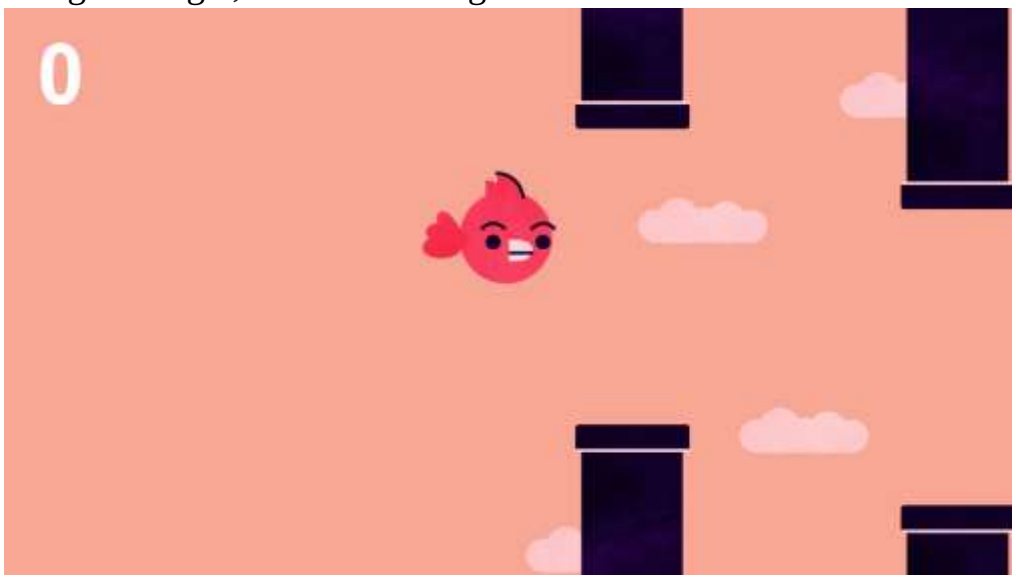


Figure 3: The Main Scene.

- **Characters**: There are three main characters in the game, as follows.
  - **The Bird**: It is the main character in the game, as shown in Figure 4. It is fully controlled by the player. The player can only tap on the space bar in the keyboard to let the bird to fly up with small vertical distance. The bird must go down again by the gravity. The bird has a wing that flips up and down repeatedly.



Figure 4: The bird.

- **The Pipes**: They are obstacles (enemies) for the player and he has to avoid them. There are endless generation for the pipes and they are moving across the screen from right to left with a specific velocity. They have two different shapes according to their orientation: up and down pipes. For each generation, one up and one down pipe must be appeared with random heights to challenge the player. All pipes that go out of the scene must be destroyed or canceled. Figure 5 shows the pipes.



Figure 5: The Pipes.

- **The Clouds**: They are a 2D sprite that will be reproduced repeatedly during the game play. They must be positioned randomly in the scene and moved across the screen from right to left. They are neutral and will not affect the game scenario. Once any cloud goes out of the scene, it must be destroyed. Figure 6 shows the Clouds.



Figure 6: The Clouds.

- **User Interface (UI)**: There are one Score text and Game Over panel, as follows.
    - **The Score**: It is a text to show the current score of the player during the game play, as shown in Figure 7. It is located in the top-left corner of the Main scene. It starts with 0 and will be incremented by 1 each time the player passes between two pipes successfully without touching them.



Figure 7: The Score.

    - **Game Over Panel**: It is a panel that indicates the end of game, as shown in Figure 8. It is hidden by default and will be shown in the middle of the Main scene only when the player touches the pipe' edge or the ground. It has one button called "Play Again" to let the user to reset the Main scene and plays the game from the beginning.

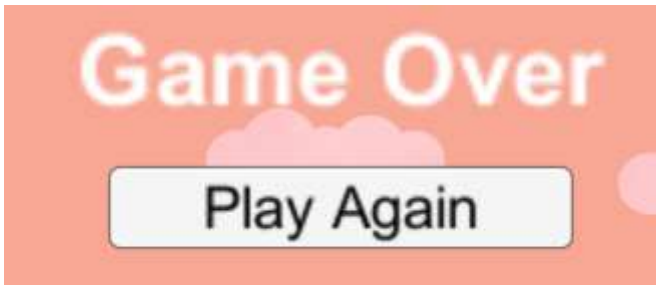


Figure 8: The Game Over Panel.

- **The Sound Effects**: There will be some sound effects during the game play to add interactivity with users. There are five short sound clips, as follows.
    - **Swoosh**: It is a sound clip that will be played at the beginning of the game.
    - **Flap**: It is a sound clip that will be played each time the player hits the space bar.
    - **Point**: It is a sound clip that will be played each time the player successfully passed through the pipes.
    - **Collision**: It is a sound clip that will be played each time the player wrongly touched any pipe's edges.
    - **Die**: It is a sound clip that will be played once the player reached the game over case.